

Auditable Driver-State Annotation from Naturalistic In-Cabin Video

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ABSTRACT

Naturalistic driving studies often rely on in-cabin video to study driver attention and supervision, yet many datasets lack synchronized eye tracking or frame-level driver-state labels. This creates a measurement gap for downstream behavioral analysis: researchers need scalable features that describe where drivers look and whether their hands remain on the steering wheel, but fully manual coding does not scale to long recordings with many participants. This work-in-progress paper presents an auditable human-in-the-loop workflow that converts ordinary in-cabin video into area-of-interest (AOI) glance categories and hand-on-wheel state variables. The workflow combines reviewed driver ROI specification, consensus AOI annotation, participant calibration, frame-level inference, temporal stabilization, hand/wheel detection, state distillation, and window-level metric extraction. We report preliminary evidence from 15 participants, including 301 processed gaze/wheel segments, about 3.91 million frame-level outputs, 7,306 coverage-qualified behavior windows, a 14-participant AOI leave-one-participant-out experiment, a few-shot adaptation curve, a temporal stabilization ablation, hand-on-wheel review subsets, a teacher-student hand-state distillation check, and ONNX deployment checks. The contribution is not a new gaze backbone or an eye-tracking replacement. Instead, the paper defines a reusable annotation and validation workflow for downstream studies that require transparent, reviewable driver-state features from naturalistic video.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → *Computer vision tasks*.

KEYWORDS

Naturalistic driving, driver monitoring, glance annotation, human-in-the-loop, area of interest, hand-on-wheel

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1 INTRODUCTION

Naturalistic driving studies preserve real traffic, cabin layouts, lighting, and driver behavior, but many in-cabin video datasets are recorded with ordinary cameras rather than synchronized eye trackers. They therefore lack fixations, point-of-regard estimates, and area-of-interest (AOI) glance labels.

Before such datasets can support questions about monitoring, distraction, takeover readiness, or automation engagement, researchers must first turn video into behavioral variables.

Manual coding is interpretable but difficult to scale because it requires coder training, consensus review, and reliability audit. Fully automated vision is attractive, but a black-box classifier is not sufficient for naturalistic driving data: a prediction can be invalid when the driver crop is wrong, a passenger appears in the selected region, a face detector fails, or short flicker is interpreted as a behavioral event.

Prior work motivates coarse regions and behavioral states rather than exact gaze or hand geometry. Video-based driver visual behavior standards define glance measures as observable AOI episodes and durations [7, 11]. Gaze-region work shows that head pose, eye-region appearance, and scene geometry can support roadway or in-vehicle glance classification without eye tracking [4, 5]. For steering engagement, recent work has studied driver hand detection, synthetic hands-on-wheel data, and in-vehicle hands on/off prediction [1, 10, 12]. Our contribution is complementary: we do not claim a new detector or an external benchmark result, but define an auditable workflow that turns ordinary naturalistic in-cabin video into documented AOI and hand-state features.

We therefore treat driver-state extraction as a measurement workflow rather than a single model score. YOLOv8-cls provides a compact train-export-deploy path [8]; SCRFD supports driver ROI review [6]; and GroundingDINO localizes text-prompted hands and steering wheels [9]. Humans still define the study geometry and semantics: they confirm driver regions, label AOIs, review outputs, and inspect uncertain hand states.

The resulting measures are AOI-level driver-state features, not continuous gaze estimates or eye-tracking substitutes [4, 7, 11]. This WIP defines the workflow, reports preliminary checkpoint evidence from 15 participants, evaluates model/deployment choices, and adds a state-distillation checkpoint for the slow hand/wheel branch before any large-scale student replacement is claimed.

2 METHOD

Figure 1 summarizes the workflow. The input is a set of target videos and analysis segments defined by a study spreadsheet. The workflow resolves local videos, assigns or reviews driver regions, constructs annotation packs, collects human AOI labels, trains participant-specific AOI classifiers, exports deployment models, runs segment-level inference, stabilizes

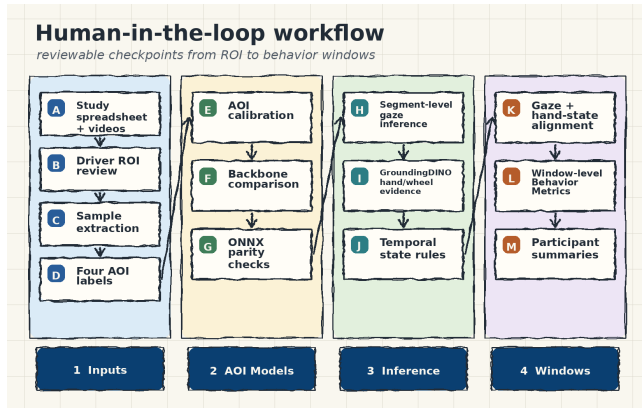


Figure 1: Human-in-the-loop workflow for deriving AOI-level glance and hand-on-wheel variables from naturalistic in-cabin video. The method keeps ROI provenance, frame semantics, temporal stability, hand-state review, and behavior aggregation as separate reviewable stages.

frame-level predictions, estimates hand-on-wheel states, optionally distills GroundingDINO teacher outputs into a light-weight state student, and aggregates the resulting streams into behavioral windows.

2.1 Measurement Targets

The gaze branch assigns each interpretable frame to one of three valid AOI classes. *Forward* denotes attention toward the forward roadway. *In-Car* denotes attention toward in-vehicle targets such as the instrument cluster, center display, mirrors, or controls. *Non-Forward* denotes attention away from both the forward roadway and in-vehicle targets. A fourth label, *Other*, is used for invalid or non-interpretable frames. Primary behavior metrics use the three valid AOI classes, while *Other* is retained for cleaning, model training, and secondary error analysis.

The hand branch estimates steering-wheel engagement as *ON*, *OFF*, or *UNCERTAIN*. These states are derived from hand and steering-wheel detections, spatial contact evidence, and temporal state rules. For window-level analysis, hand-state samples are aligned with the gaze stream so that researchers can compute both overall and AOI-conditioned hands-on/off ratios.

2.2 Human ROI and AOI Annotation

ROI selection is part of the measurement method, not an invisible preprocessing step. Reviewers record simple driver ROIs directly; in dual-panel or multi-view layouts, face evidence ranks candidate panels but the selected panel remains visually confirmed. Uncertain assignments export grid-overlay references and fill-in manifests for manual correction.

After ROI specification, representative frames are sampled across each target video and coded as *Forward*, *In-Car*,

Non-Forward, or *Other*. The *Other* class captures systematic invalid samples from occlusion, lighting, camera layout, or non-driver faces, preventing these frames from contaminating valid AOI metrics.

The annotation process preserves consensus decisions while keeping reliability audit as an explicit WIP activity. Coders are trained on a shared manual and exemplar frames, production labels are reviewed before consensus, and disagreements are resolved through joint discussion. Chance-corrected reliability indices such as Cohen’s κ [2] are useful when a completed double-coded audit set is available; Fleiss’ κ is appropriate only when more than two coders independently label the same items [3]. Because the present WIP is still in the feature-construction stage, we treat aggregate agreement as descriptive process evidence and leave the formal chance-corrected reliability audit for the next validation pass.

2.3 AOI Classification and Temporal Stabilization

The operational AOI branch uses a YOLOv8-cls classifier because it provides a compact training and deployment path. Participant-specific calibration starts from a small set of labeled samples and produces a participant-adapted classifier for segment-level inference. At inference time, SCRFD detects faces within the reviewed driver ROI; a layout-specific target-face rule rejects implausible identity jumps and selects the driver face; the selected crop is then classified into the four AOI labels.

Deployment convenience does not by itself establish measurement validity. For that reason, the repository also includes a model-comparison workflow that trains YOLOv8-cls, ResNet, EfficientNet, ConvNeXt, and ViT backbones under a common participant holdout protocol. The comparison is not meant to find a universal best backbone; it tests whether the deployment choice has an irreplaceable accuracy advantage under generalization and calibration constraints.

Frame-level outputs are not used directly as behavioral events. Presence hysteresis prevents brief missed detections from producing invalid states, and class debounce requires a candidate AOI label to persist before replacing the current stable label. This keeps frame semantics and temporal event structure analytically distinct.

2.4 Hand-on-Wheel State Estimation

The hand branch uses the steering-wheel camera view. GroundingDINO detects hands and steering wheels with text prompts. Contact evidence is computed from overlap between detected hand boxes and steering-wheel boxes. The frame-level state machine maps reliable contact evidence to *ON* or *OFF*; unreliable frames retain the previous state for a short grace period before entering *UNCERTAIN*. A temporal majority rule then produces the stable state sequence used for review and metric extraction.

GroundingDINO is useful for text-prompted bootstrapping but too slow for repeated large-scale processing, so it is treated as a teacher detector. Sampled teacher detections

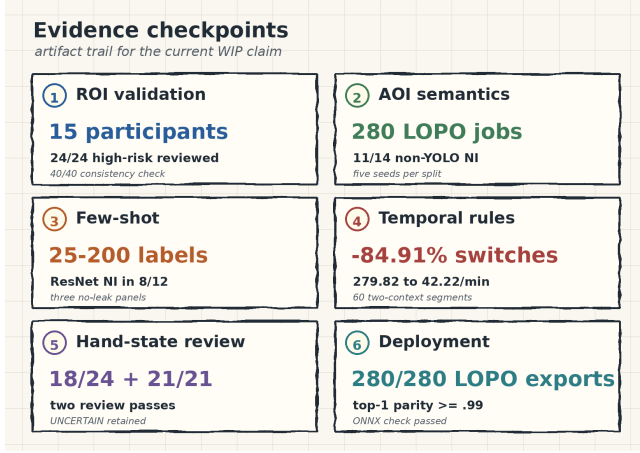


Figure 2: Evidence checkpoints used to evaluate the annotation workflow. Each checkpoint corresponds to a distinct source of possible measurement error.

can supervise either a two-class hand/wheel detector student or, as used here, a state-level ROI-crop classifier supervised by stable *ON/OFF/UNCERTAIN* outputs. This branch is reviewed at the final-state level, after detector confidence, contact rules, and temporal stabilization act together.

2.5 Window-Level Metrics

The stabilized gaze and hand-state streams are aggregated into fixed analysis windows. The window is not a new labeling task; it is a deterministic, timestamp-weighted aggregation of the stabilized AOI and hand-state streams. This design reduces the influence of short-term prediction flicker, matches the downstream 20 s analysis unit, and keeps each derived value traceable to frame-level states, valid denominators, coverage gates, and episode-duration rules. Let g_i be the stabilized AOI label at time t_i and let $\mathcal{L} = \{Forward, In-Car, Non-Forward\}$ be the valid AOI set. Samples labeled *Other* are excluded from valid-AOI denominators. AOI time percentages use timestamp-based frame-duration weights rather than raw frame counts.

Glance counts are computed as entries into AOI episodes. Consecutive frames with the same valid AOI label are treated as one episode. Long off-path glances are defined as consecutive frames in either *In-Car* or *Non-Forward*, with event counts reported at 1.6 s and 2.0 s thresholds. Hand-on-wheel ratios are computed overall and conditioned on AOI label after excluding *UNCERTAIN* and unmatched wheel-state samples. Windows failing the gaze-coverage threshold are kept for QC but excluded from participant-level summaries.

3 EXPERIMENTS

The evaluation focuses on whether the workflow produces reviewable measurement artifacts, not whether it solves driver monitoring in a single end-to-end score. Unless otherwise stated, evidence reflects artifacts finalized by May 24–25, 2026 for the 15-participant analysis set.

Table 1: Processing coverage across the 15-participant analysis set.

Coverage item	Value
Participants in analysis	15
Matched target videos	176
Gaze segments with CSV outputs	301
Wheel segments with CSV outputs	301
Frame-level gaze rows	3,914,416
Frame-level wheel-state rows	3,914,412
Generated 20 s windows	7,327
Coverage-qualified windows	7,306
QC-excluded windows	21

Table 2: Mean held-out performance in the five-seed 14-participant LOPO AOI experiment. Each model has 70 held-out runs.

Model	Macro-F1	Bal. acc.	Event acc.
ResNet50	0.596	0.625	0.627
EfficientNet-B0	0.575	0.601	0.616
YOLOv8s-cls	0.574	0.606	0.618
DeiT-Tiny	0.300	0.350	0.399

3.1 ROI and Processing Coverage

The ROI branch covers 15 participants with human-specified or human-confirmed ROI assignment files and production gaze/wheel ROI manifests. The completed audit covered provenance, resolution, visual validity, uncertain dual-ROI rows, low-margin detector-assisted rows, and layout-specific gaze/wheel checks. All 24 high-risk rows were resolved, and a manual/current ROI consistency check matched 40/40 rows with zero coordinate changes.

The rebuilt inference plans, gaze maps, wheel maps, and 20 s behavior windows cover all 15 analysis participants. Table 1 summarizes the resulting processing coverage. All 301 planned gaze/wheel segments had frame-level CSV outputs, and all 7,327 generated windows resolved to gaze and wheel. The final participant-level summaries used 7,306 windows passing the 0.98 gaze-coverage threshold; 21 windows were retained for QC only.

3.2 AOI Generalization and Adaptation

To evaluate cross-participant generalization, we ran a 14-participant leave-one-participant-out (LOPO) AOI experiment with five seeds and four backbones, for 280 training runs. The primary metric is three-class macro-F1 over *Forward*, *In-Car*, and *Non-Forward*; *Other* remains a secondary data-quality label because current test sets have limited invalid-sample support.

All training and prediction jobs completed. Table 2 reports mean held-out performance across the 70 runs per model. ResNet50 achieved the highest mean metrics; YOLOv8s-cls remains useful for deployment but was not uniquely accurate.

At the split level, the best non-YOLO mean exceeded YOLOv8s-cls on 11/14 participants. A 3 percentage-point bootstrap non-inferiority test found at least one non-YOLO

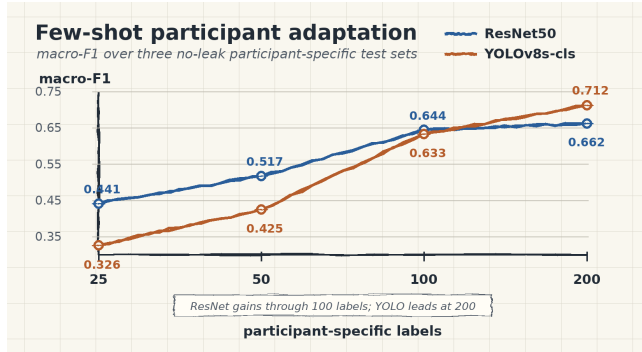


Figure 3: Few-shot no-leak adaptation curve over three participant-specific adaptation panels (five seeds per participant-budget cell).

candidate satisfying the margin on 11/14 splits, supporting family-best non-inferiority rather than exact equivalence.

The few-shot experiment tests how much participant-specific labeling is needed when the goal is adaptation rather than zero-label generalization. We ran a no-leak adaptation curve over three participant-specific adaptation panels at 25, 50, 100, and 200 label budgets, five seeds per budget, and frozen participant-specific test sets. Figure 3 shows rapid ResNet50 gains from 25 to 100 labels and YOLOv8s-cls leading at 200 labels. ResNet50 was non-inferior to YOLOv8s-cls in 8/12 participant-budget cells, suggesting that label budget should be controlled and reported.

3.3 Temporal Stability

We compared raw per-frame AOI predictions with stabilized *Gaze_Class* on a two-context subset covering 60 segments, 643,603 frames, and 436.95 min. Stabilization reduced label switches from 279.82 to 42.22 per minute, an 84.91% reduction. Raw and stable labels differed on 106,418 frames, or 16.54% of the subset. Stabilization also changed behavior-level event structure: 1.6 s off-path episodes rose from 923 to 1,289, and 2.0 s episodes rose from 696 to 1,032, showing that temporal rules change event structure rather than cosmetic smoothness alone.

3.4 Hand-State Review and Distillation

The wheel branch produced 301 segment-level outputs and 3,914,412 state-classified rows. Several driver/camera contexts had majority-*UNCERTAIN* support, indicating that hand-state estimation is sensitive to camera layout, occlusion, and steering-wheel evidence. AOI/glance metrics can be used independently; hand-conditioned metrics should report the valid *ON/OFF* denominator, and majority-*UNCERTAIN* contexts should be treated as low-support hand evidence.

The first agreement summary contained 24 reviewed rows with matched pipeline and human labels. Agreement was 18/24 (75.0%), with four false-ON and two false-OFF cases. During review, we found that one evidence batch initially reused single-panel gaze ROIs as wheel ROIs. We therefore

Table 3: Hand-on-wheel final-state validation outcomes.

Scope	Reviewed	Agreement	False ON/OFF
Initial manifest	24	18/24	4/2
Corrected second pass	21	21/21	0/0

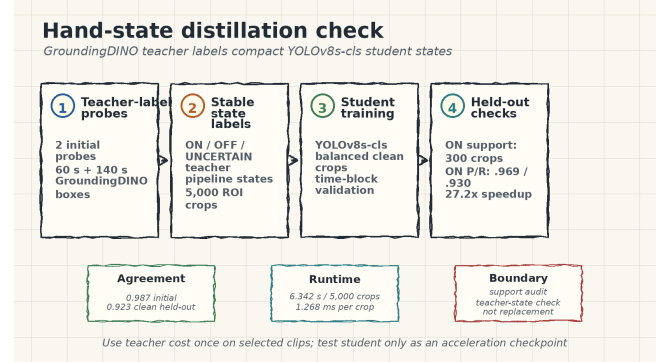


Figure 4: State-distillation check for the hand-on-wheel branch. GroundingDINO is treated as a teacher on selected probes; the compact student is evaluated as an acceleration checkpoint rather than as a replacement for full multi-participant validation.

regenerated layout-specific cockpit evidence for the affected participants. The corrected second pass covered 21 final states; the reviewer matched all 21 pipeline states, including 12 binary *ON/OFF* cases and nine *UNCERTAIN* cases (Table 3). We interpret this second pass as a post-correction workflow sanity check showing that the regenerated evidence is internally consistent with reviewed final states, not as an independent human-gold-standard validation of hand-state measurement.

We ran a bounded state-distillation check for the slow hand/wheel branch. An initial two-probe runtime check, 60 s and 140 s, produced 5,000 ROI crops labeled by the teacher pipeline’s stable *ON/OFF/UNCERTAIN* state; a deterministic frame-hash split held out 1,035 validation crops. Because that split had few *ON* samples, we added an *ON*-enriched multi-context time-block check using clean teacher frames with stable/raw agreement, at least 0.5 s separation from stable-state transitions, reliable contact evidence for *ON*, and balanced held-out support of 300 validation crops per state. This WIP check reports teacher-state agreement and per-class support; it does not claim that the student can replace the teacher branch in production.

Table 4 summarizes the checks. The initial saved student matched 1,022/1,035 teacher states (0.987 agreement) and processed 5,000 crops in 6.342 s after warm-up, versus 172.752 s for the two GroundingDINO probes (27.2× speedup), but its *ON* support was only 12 crops. The *ON*-enriched clean time-block check raised *ON* support to 300 held-out crops and reached 0.969 precision, 0.930 recall, and 0.949 F1 for

Table 4: Teacher-student hand-state distillation checks.

Check	State	N	P	R	F1
Init.	All	1,035	0.987	agree.; 27.2× speedup	
Init.	OFF	540	0.993	0.983	0.988
Init.	ON	12	0.857	1.000	0.923
Init.	UNC.	483	0.986	0.992	0.989
Clean	All	900	0.923	agree.; support audit	
Clean	OFF	300	0.894	0.897	0.895
Clean	ON	300	0.969	0.930	0.949
Clean	UNC.	300	0.910	0.943	0.926

Table 5: Compact ONNX CPU deployment benchmark from the holdout protocol.

Model	MB	B1 p50 ms	B1 img/s	B32 img/s
YOLOv8n-cls	5.5	4.85	202.2	1621.2
YOLOv8s-cls	19.4	5.26	183.7	903.6
ResNet50	89.6	7.20	139.1	525.7
DeiT-Tiny	21.2	17.19	56.7	291.1
EfficientNet-B0	15.3	18.63	53.4	288.3
ConvNeXt-Tiny	106.2	18.65	51.8	114.1
EfficientNet-B3	40.8	33.69	29.4	157.5

that class. The overall clean held-out agreement was 0.923, with *OFF* requiring additional boundary-context review. We therefore treat the student as an acceleration checkpoint and support audit, not as a production replacement for the teacher branch.

3.5 Deployment Checks

Deployment checks support engineering feasibility. In the five-seed LOPO matrix, all 280 trained models exported to ONNX and no top-1 parity result fell below 0.99. Table 5 reports a compact CPU benchmark from the holdout protocol, where all 120 exports completed. YOLOv8n and YOLOv8s are smaller and faster than non-YOLO backbones, supporting YOLOv8-cls as a convenient but not uniquely accurate deployment path.

4 DISCUSSION

The central design choice is audibility: ROI selection, AOI semantics, participant calibration, temporal stability, hand-state review, and window aggregation remain separate so errors can be localized rather than hidden inside an aggregate model score. The evidence also clarifies that deployment convenience is not unique accuracy. YOLOv8-cls is compact and exportable, but ResNet50 and EfficientNet-B0 remain competitive for AOI generalization/adaptation; for hand state, a compact student is useful only as a support audit until broader human-labeled, multi-context validation is complete.

For downstream studies, the value is citation-level provenance. A later analysis can cite this workflow as the source of AOI glance and hand-on-wheel feature construction while keeping its own claims focused on behavioral questions, not feature-construction mechanics.

5 LIMITATIONS AND OPEN WORK

Several limitations remain. The workflow depends on participant labels and is not a zero-label generalization system. High *Other* rates can reduce valid-AOI denominators and should be reported downstream. ROI or layout errors can dominate results, so new layouts, videos, or manifest edits require repeated review. The hand-on-wheel second pass is single-reviewer evidence and should be expanded into an independently double-coded validation set. Window aggregation reduces short-term flicker and makes behavior summaries auditable, but it does not remove classifier-derived uncertainty in thresholded episode counts such as long off-path glances. Current distillation evidence is teacher-state agreement, not full human-labeled multi-participant validation; the student must be stress-tested before replacing GroundingDINO. Deployment benchmarks should be repeated on target hardware before strong latency claims.

6 CONCLUSION

This WIP presents a human-in-the-loop workflow for extracting AOI-level glance and hand-on-wheel variables from naturalistic in-cabin video. Across 15 participants, the workflow processed millions of frames while retaining reviewable checkpoints for ROI validity, AOI generalization, participant adaptation, temporal stability, hand-state plausibility, student acceleration, and deployment feasibility. This result is possible because the pipeline keeps geometry selection, human AOI semantics, frame-level inference, temporal smoothing, hand-state review, and window-level aggregation as separate auditable stages rather than merging them into a single model score. The evidence therefore supports AutoUI as a transparent feature-construction method for downstream naturalistic driving studies, not as an eye-tracking replacement, a production hand-state detector, or a standalone theory of driver behavior.

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